Centre Informatique du Ministère de la Santé (CIMS)

Internship Report

Development of a Tumor Detection Software using Deep Neural Networks for MRI Analysis

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Internship Period: from 24/06/2024 to 24/08/2024

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Supervisor: Mr. Akrem Allani

September 25, 2024

Signature of Supervisor: ______________________

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Acknowledgments

Before detailing the development of my experience during this internship, it is important to start this report with thanks to those who welcomed me, shared their knowledge, and made this internship a highly rewarding experience.

First and foremost, I would like to express my sincere gratitude to **Mr.Akrem Allani** and **Mr.Lotfi Allani** for accepting me as an intern and showing remarkable kindness throughout my time here. I am especially grateful for their guidance and support throughout this internship, allowing me to gain valuable knowledge and experience in the field.

I would also like to extend my heartfelt thanks to the entire team for making this internship a truly enjoyable experience. Every day was a pleasure to work alongside passionate professionals, and I greatly appreciated the opportunity to learn from them and grow in such a supportive environment.

General Introduction

In recent years, deep learning has significantly advanced medical imaging, particularly in the detection and classification of brain tumors. A brain tumor, whether benign or malignant, can cause severe complications due to its location in the confined space of the skull. Early and accurate detection is crucial for effective treatment. In this project, I developed a software solution using Convolutional Neural Networks (CNNs) to classify brain tumors based on MRI images. The system utilizes TensorFlow 2.16, leveraging its enhanced data augmentation and optimized performance for fast, accurate tumor detection.

The dataset used includes 7,023 MRI images classified into four categories: **Glioma**, **Meningioma**, **Pituitary Tumor**, and **No Tumor**. The CNN model helps automate tumor detection, potentially improving diagnostic accuracy and speed in neurology. Furthermore, I deployed the model as a web application using Django, allowing healthcare professionals to upload MRI scans and receive real-time tumor classification results. This project showcases the potential of machine learning in improving healthcare outcomes.

Chapter 1

Company and Project Presentation

1.1 Company Presentation

1.1.1 Industry Sector

The Centre Informatique du Ministère de la Santé (CIMS), created by Law No. 92-19 on February 3, 1992, is a public institution with a non-administrative character under the Ministry of Health in Tunisia. The CIMS plays a major role in the digitization of the public health sector through the development of e-health solutions.

As a key player in the field of digital health, CIMS is involved in the design, development, and deployment of hospital information systems, as well as the implementation of digital services intended for public healthcare structures.

1.1.2 Objectives and Mission of the Company

The main missions of CIMS are:

- **Development and management of hospital information systems**: This includes the design and implementation of digital tools for managing hospital processes.
- **Support and training for public healthcare structures in their digital transitions**: CIMS provides technical assistance and continuous training to hospital and health center staff to help them effectively use information systems.
- **Security and compliance of information systems**: CIMS ensures that the digital tools implemented comply with current security regulations, thus guaranteeing the protection of sensitive data for patients and healthcare institutions.

By playing an essential role in the digitization of the healthcare sector in Tunisia, CIMS contributes to improving the quality of care services by facilitating the adoption of digital technologies within public healthcare institutions.

1.2 Internship Context and Problem Statement

This internship project is part of the current trend where automatic analysis of medical images using neural networks has become an essential tool for improving diagnostic accuracy. The internship's goal was to develop software capable of detecting brain tumors from MRI images based on convolutional neural networks (CNNs).

1.3 Project Objectives

The main objective of this project is to develop an AI-based solution for the automatic detection of brain tumors from MRI images, aiming to enhance the accuracy and speed of diagnoses in public healthcare structures in Tunisia. The project specifically addresses the needs of healthcare professionals by providing an accessible, reliable tool that complies with medical data security standards.

The specific objectives of the project are as follows:

- **Adaptation of a pre-trained tumor detection model:**
  - Utilize a pre-existing Convolutional Neural Network (CNN) model available on Kaggle and adapt it to improve its performance.
  - Optimize the model's hyperparameters and layers to achieve better accuracy on MRI brain tumor images.
  - Apply data augmentation techniques to enhance the model's robustness against the variability in the available MRI images.
- **Development of a user-friendly web interface:**
  - Develop a web application using the Django framework to allow users (such as doctors and radiologists) to upload MRI images and receive predictions on the presence or absence of tumors.
  - Implement a simple and intuitive interface with clear features, such as image uploading and prediction result display.
- **Deployment in a production environment:**
  - Integrate the model and web application into a secure cloud environment for easy access by end-users.
  - Ensure that the application is scalable and optimized for production use, particularly through the use of Docker containers for system portability and reliability.
  - Ensure compliance with medical data security and privacy standards to protect patient and healthcare institution information.
- **Performance evaluation and results:**
  - Analyze and compare the model's performance on MRI images using metrics such as accuracy, recall, precision, and F1-score.
  - Conduct thorough testing to ensure that the model operates optimally in a production environment with real-world data.
- **Documentation and training:**

- Write comprehensive technical documentation on the model adaptations and the functioning of the web application.
- Provide a user guide for healthcare professionals to help them fully utilize the diagnostic tool.

These objectives aim to deliver a complete solution that improves patient care for those with brain tumors while contributing to the digital transformation of public healthcare structures in Tunisia.

Chapter 2

Work Environment Study

2.1 Hardware Environment and Infrastructure

The hardware used for this project included:

- Personal computer:
  - **Processor**: Intel Core i7
  - **RAM**: 32 GB
  - **Graphics card**: NVIDIA GeForce GTX 1660
- **Google Colab** for training the model with a GPU.
- **Visual Studio code** for training the model with a GPU.

2.2 Software Environment and Technologies Used

The software tools used in this project include:

- **TensorFlow 2.16** for training the CNN model.
- **Django** for developing the web interface.
- **Python** for training scripts and data processing.
- **Docker** for containerizing the application.

2.3 System Architecture

The system consists of three components:

- A trained CNN model to detect tumors in MRI images.
- Django backend to handle requests and model calls.

Chapter 3

Dataset Description

3.1 Overview of the Dataset

The dataset used for the tumor detection task comprises **7,023 brain MRI images**. These images were sourced from three primary datasets: *Figshare*, *Br35H*, and an additional dataset that was excluded due to low quality. The data has been organized into four classes: **Glioma**, **Meningioma**, **Pituitary**, and **No Tumor**, allowing the model to differentiate between various types of tumors and normal brain scans.

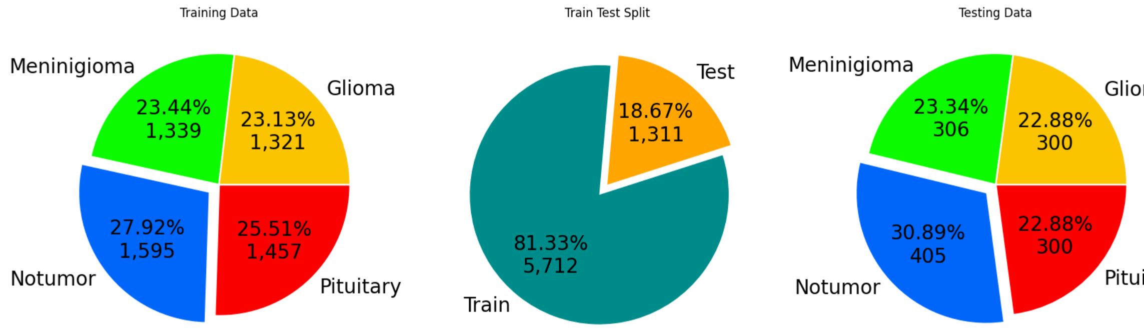

Figure 3.1: Dataset Distribution across Glioma, Meningioma, Pituitary, and No Tumor categories.

3.2 Dataset Classes and Characteristics

The four classes in the dataset are defined as:

- **Glioma**: Cancerous tumors originating in glial cells.
- **Meningioma**: Non-cancerous tumors originating in the meninges.
- **Pituitary**: Tumors affecting the pituitary gland, which can be either cancerous or non-cancerous.
- **No Tumor**: Normal brain MRI scans with no visible tumors.

3.3 Data Preprocessing Steps

To prepare the dataset for training, several preprocessing steps were applied:

- **Resizing**: All images were resized to *168x168* pixels for uniform input size.
- **Grayscale Conversion**: The MRI images are in grayscale, simplifying image data processing.
- **Data Augmentation**: Techniques such as random flips, rotations, and zooms were applied to increase the variety of data available for training.

```
data_augmentation = Sequential([
    RandomFlip("horizontal"),
    RandomRotation(0.02, fill_mode="constant"),
    RandomContrast(0.1),
    RandomZoom(height_factor=0.01, width_factor=0.05),
    RandomTranslation(height_factor=0.0015, width_factor=0.0015,
   fill_mode="constant"),
])
```

- **Normalization**: Pixel values were normalized to a range between 0 and 1, improving model training efficiency.

```
def preprocess_train(image, label):
    # Apply data augmentation and Normalize
    image = data_augmentation(image) / 255.0
    return image, label
    ])
```

Chapter 4

Project Development

4.1 Analysis of the Base Model (Kaggle) and Modifications Made

The base model from Kaggle was adapted to meet the specific needs of the project, including adjusting hyperparameters and applying data augmentation techniques to improve robustness.

4.2 Development of the Tumor Detection Model

The model uses several convolutional layers to extract features from MRI images. These features are then used to classify the images into four categories: glioma, meningioma, no tumor, and pituitary tumor.

4.3 Implementation and Development of the User Interface with Django

The user interface, developed with Django, allows doctors to upload MRI images and receive a diagnosis within seconds. This simple interface is designed to be intuitive and user-friendly.

4.4 Training, Optimization, and Evaluation of the Model's Performance

4.4.1 Training the Model

The training was carried out on a dataset containing 7023 MRI images, divided into four classes. Several data augmentation techniques were applied, including random flipping, rotation, and zooming, to enhance model robustness. The model was trained over multiple epochs with early stopping and learning rate reduction to optimize performance.

4.4.2 Evaluation Methods

The following methods were used to evaluate the model's performance:

- **Accuracy**: The ratio of correctly predicted tumor classes to the total number of images.
- **Loss**: The categorical cross-entropy loss was calculated for both training and validation sets to monitor the model's convergence.
- **Confusion Matrix**: A confusion matrix was plotted to show the performance of the classification model by comparing the true labels with the predicted labels.
- **Precision, Recall, F1-Score**: These metrics were computed to further evaluate the model's performance on a per-class basis.

4.4.3 Results Visualization

Training and Validation Accuracy and Loss

The accuracy and loss were monitored throughout the training process, and the results are shown in the plots below. These figures depict the progression of the model's performance as training continued, helping to identify potential overfitting or underfitting.

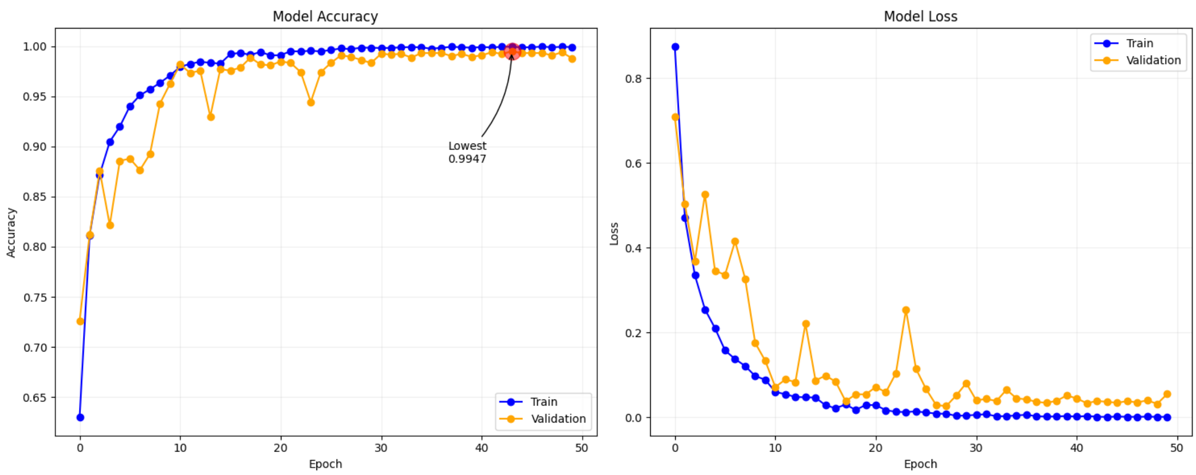

Figure 4.1: Training and Validation Accuracy and Loss

Confusion Matrix

To further evaluate the classification performance, a confusion matrix was generated to visualize how well the model differentiates between the four tumor classes.

The confusion matrix helps to pinpoint areas where the model struggles by showing where predictions were incorrect. For instance, a model might confuse similar tumor types like glioma and meningioma.

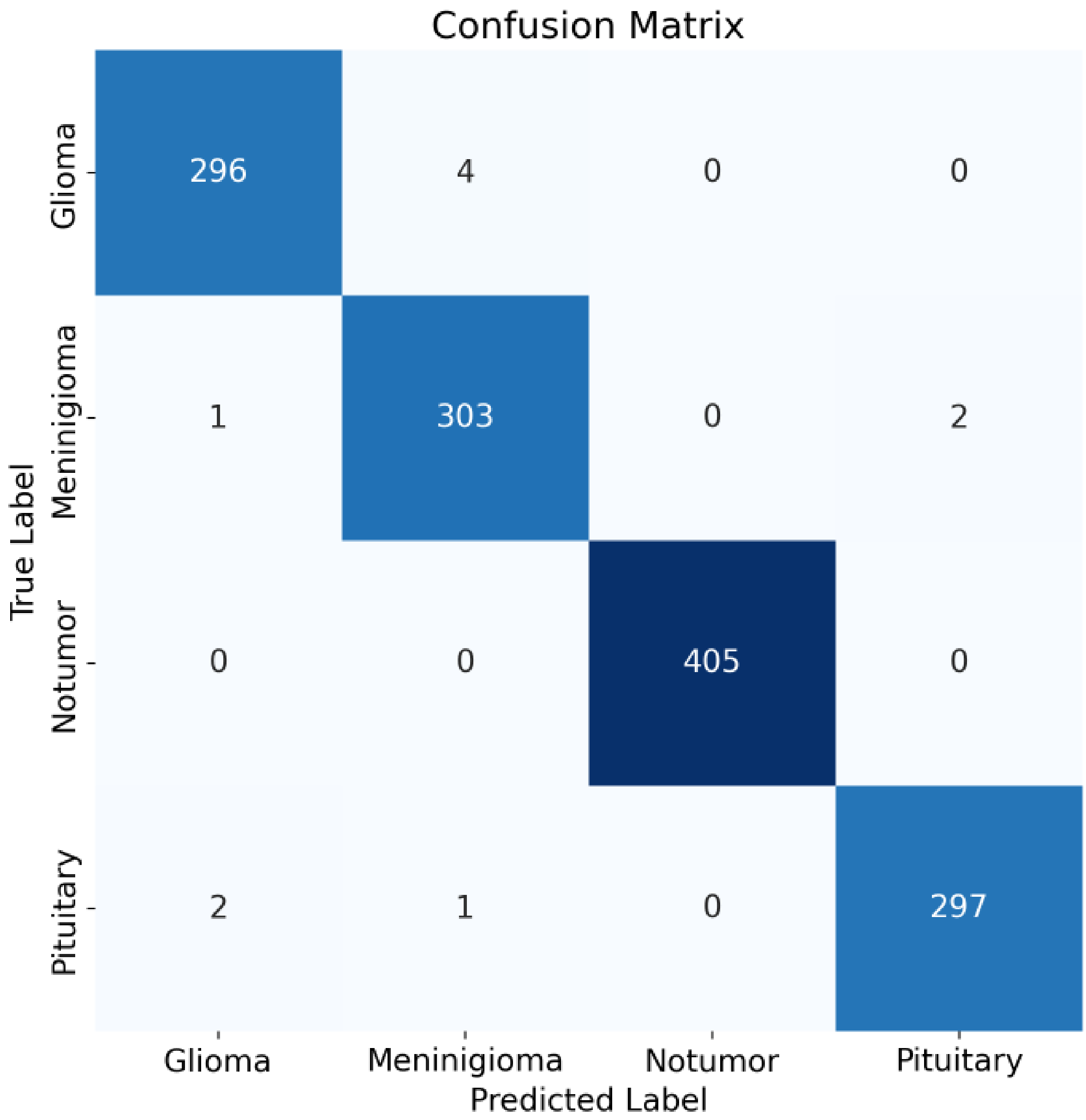

Figure 4.2: Confusion Matrix for the Model's Predictions

Class-wise Performance Metrics

Finally, precision, recall, and F1-score were calculated for each class to provide a more granular understanding of the model's performance. These metrics can be seen in the following figure:

These metrics provide insights into how well the model can identify each tumor class, highlighting the balance between precision and recall for each category.

```
Class-wise metrics:
Class: Glioma
Precision: 0.9900
Recall: 0.9867
F1-Score: 0.9883

Class: Meninigioma
Precision: 0.9838
Recall: 0.9902
F1-Score: 0.9870

Class: Notumor
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Class: Pituitary
Precision: 0.9933
Recall: 0.9900
F1-Score: 0.9917

Overall Accuracy: 0.9924
```

Figure 4.3: Class-wise Precision, Recall, and F1-Score

Chapter 5

Deployment and Testing

5.1 Web Application Deployment

The application was deployed using Django as the backend framework, rather than using a cloud platform like Heroku. The model was integrated into the Django web application, allowing users to upload MRI brain images for real-time classification.

The deployment was made on a local server, ensuring a smooth and secure interaction with the web interface. The web application includes an image upload feature, where users can select an MRI image, and the system predicts the tumor type (Glioma, Meningioma, No Tumor, Pituitary). The model used for classification was a pre-trained Convolutional Neural Network (CNN), modified for this purpose.

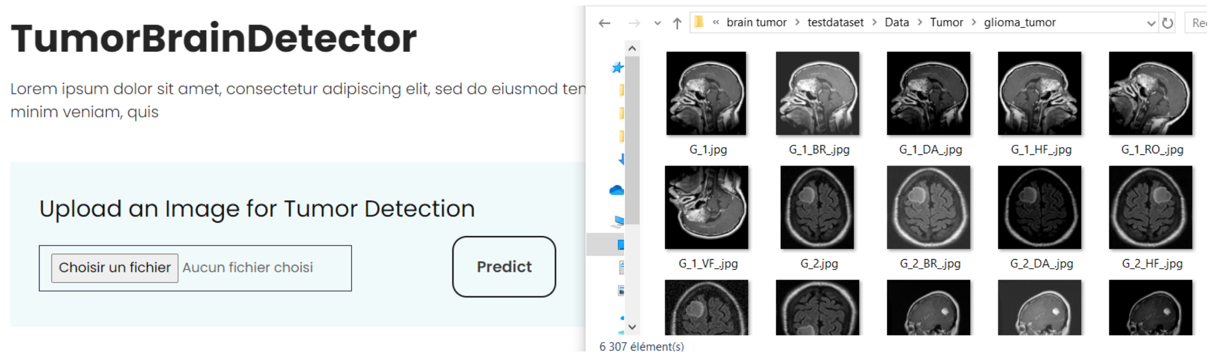

Figure 5.1: TumorBrainDetector

The TumorBrain Detector provides an intuitive interface for uploading MRI images and detecting brain tumors in real-time. Users simply select an image using the file input feature and submit it for prediction. Upon submission, the system processes the image and returns the predicted tumor class (Glioma, Meningioma, Pituitary, or No Tumor). If a tumor is detected, a warning icon is displayed alongside the result, while a checkmark icon confirms when no tumor is found, making it easy for healthcare professionals to interpret the results quickly. Below are the images showcasing the tumor detection interface and the predicted results:

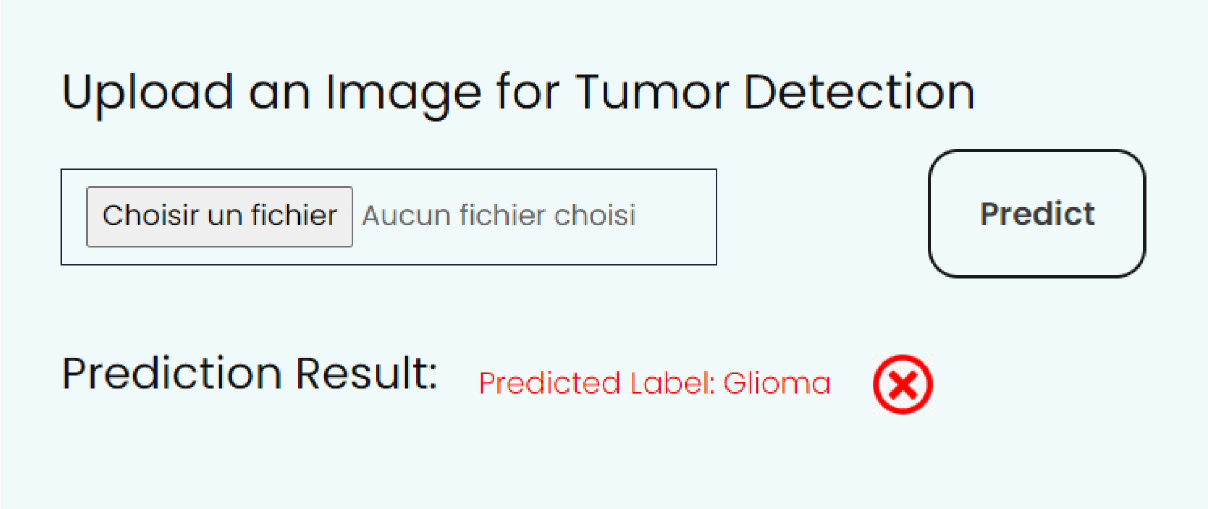

Figure 5.2: Tumor found

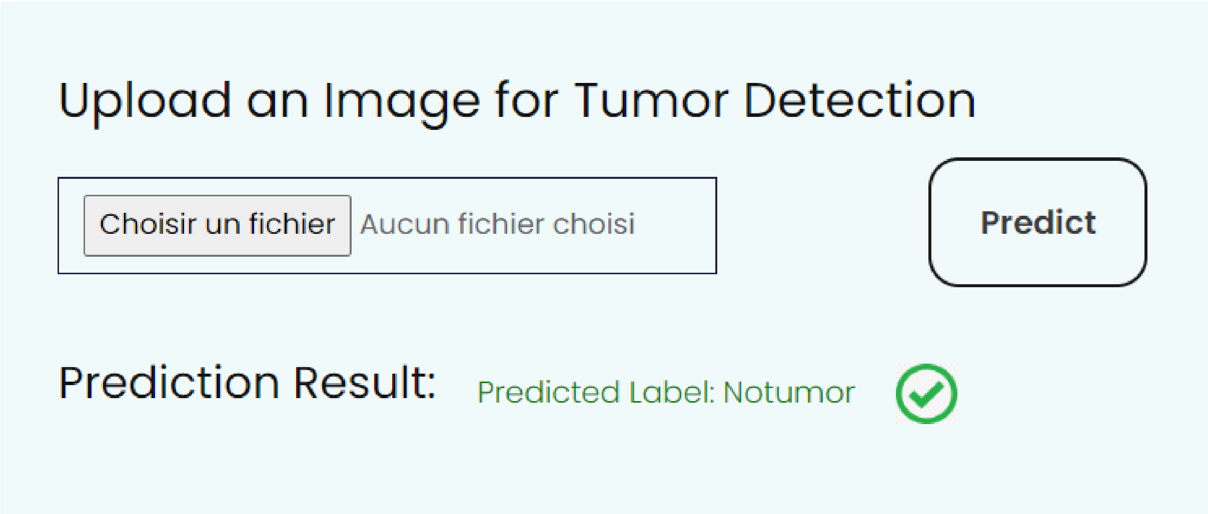

Figure 5.3: Tumor Not Found

5.2 Functional Testing and Adjustments

Functional testing was conducted to ensure the system could handle a variety of MRI images, considering different resolutions and quality levels. The tests focused on verifying the robustness of the image preprocessing pipeline and the correctness of the predictions.

Various adjustments were made to enhance the user interface by simplifying the image upload and classification process. For example, error handling was added to manage invalid image formats, and latency was reduced by optimizing the image preprocessing function..

5.3 User Interface and Template Design

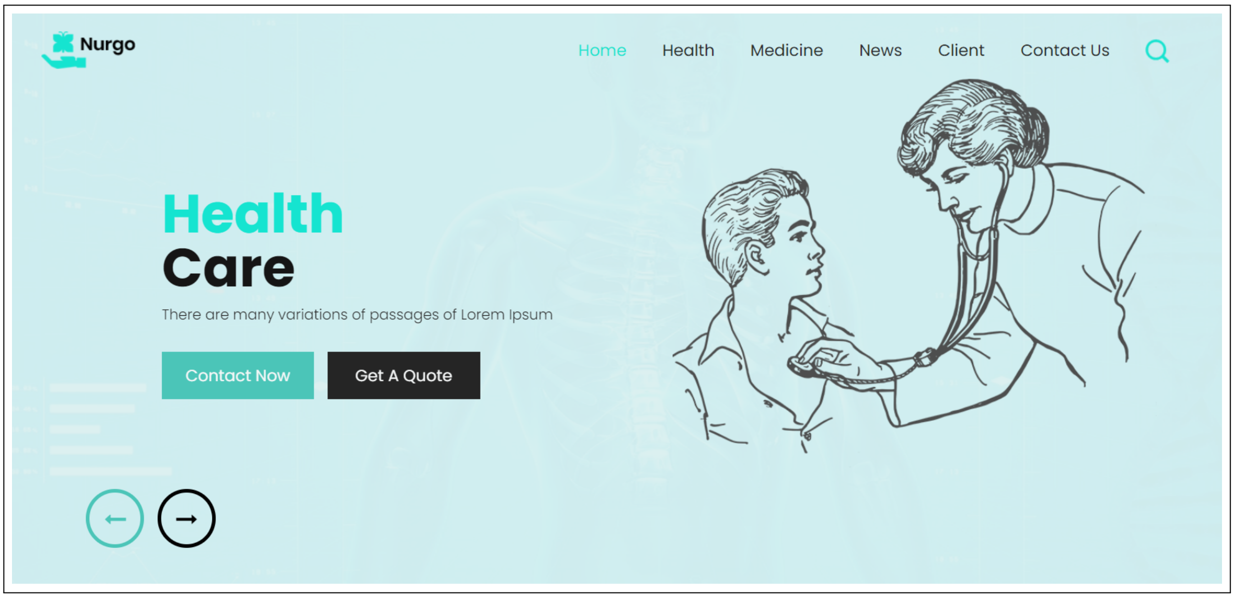

Figure 5.4: Header Section

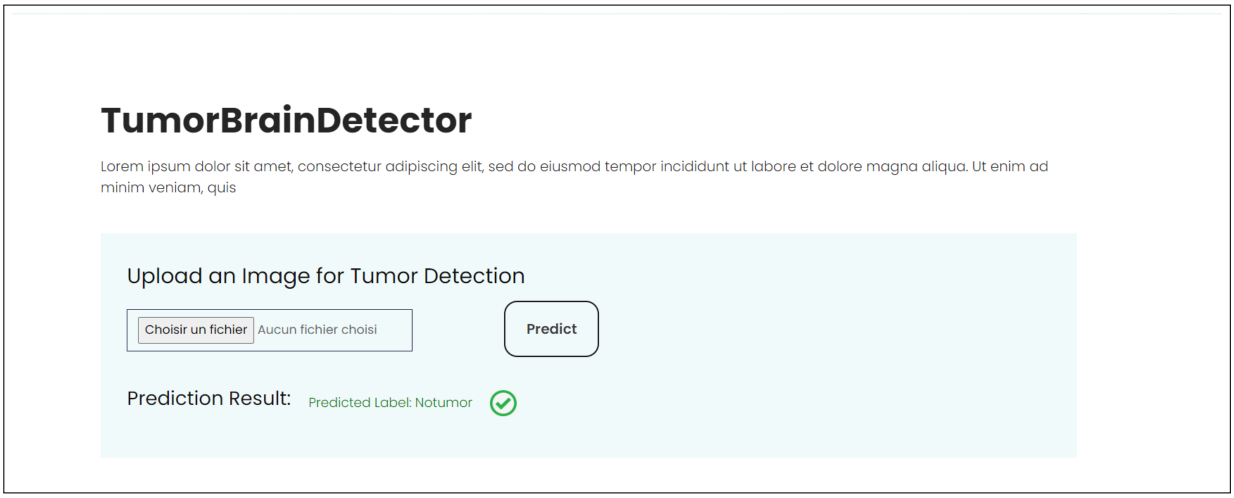

Figure 5.5: TumorBrainDetector Section

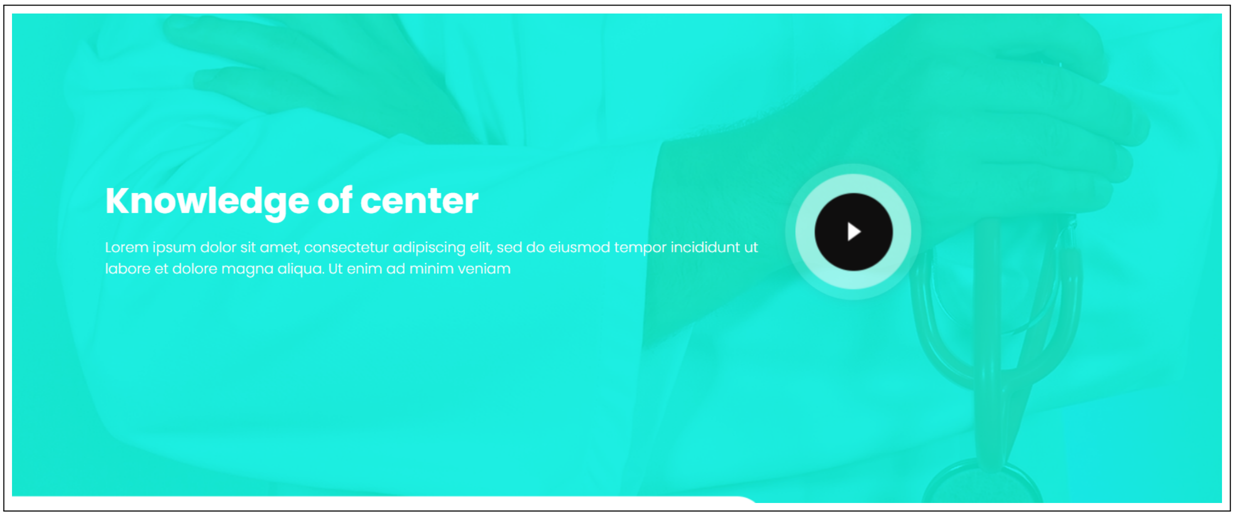

Figure 5.6: Knowledge Of Center Section

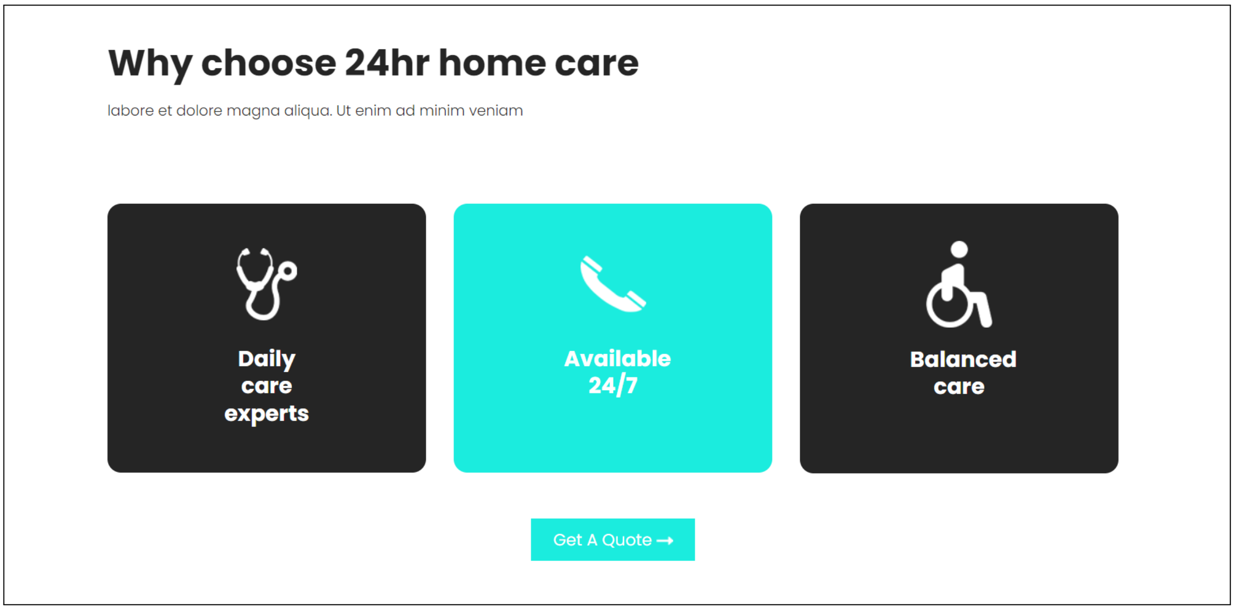

Figure 5.7: Contact Section

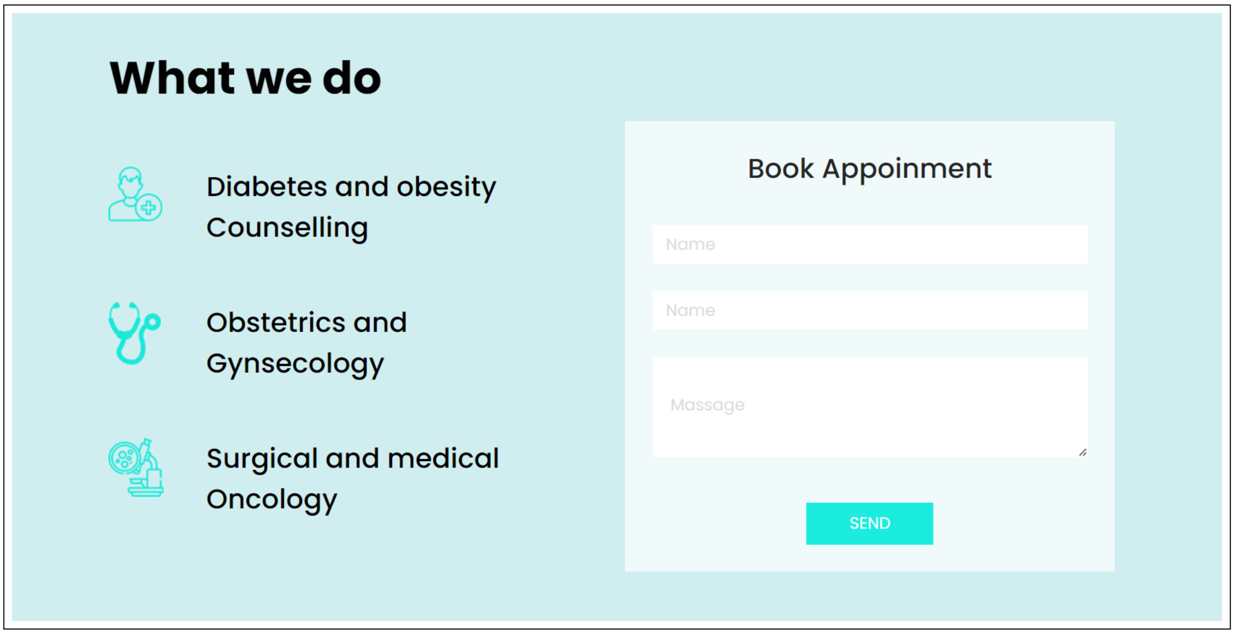

Figure 5.8: appointment Section

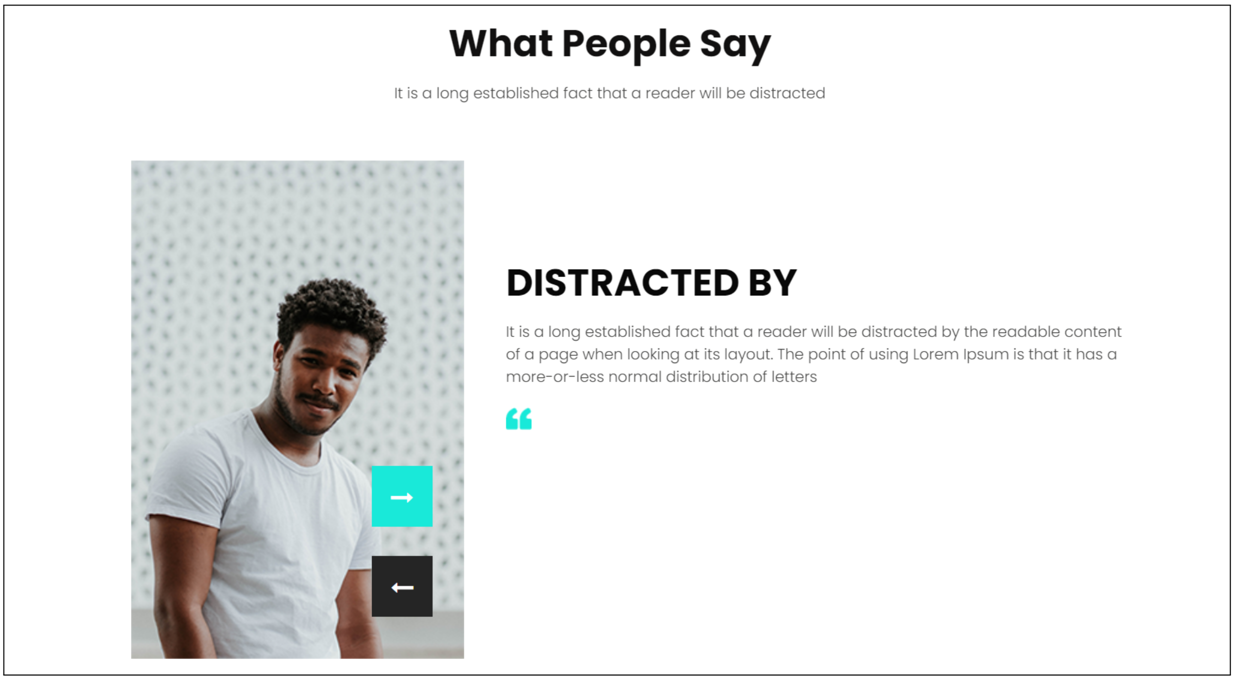

Figure 5.9: What people Say Section

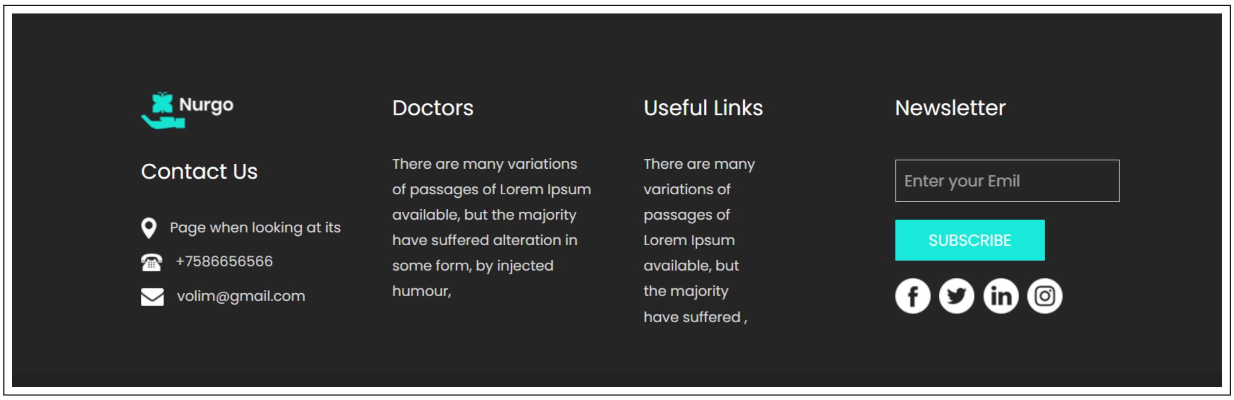

Figure 5.10: Footer Section

Chapter 6

Conclusion and Perspectives

6.1 Results Obtained and Analysis

This project successfully developed an efficient and accurate tool for detecting brain tumors from MRI images. The results obtained are promising, and the tool demonstrated good accuracy in production.

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6.2 Future Improvements and Evolution

Future improvements include integrating other types of tumors and expanding the application to analyze other medical image types, such as CT scans and X-rays. Additionally, the following enhancements are planned:

- **Add a Medical Chatbot**: Implement a medical chatbot powered by natural language processing (NLP) to assist users in navigating the application, understanding the results, and providing general information on tumor types and treatment options.
- **Enhance Model Accuracy**: Further optimize the CNN model by experimenting with additional layers, fine-tuning hyperparameters, and exploring more advanced architectures such as transfer learning or attention mechanisms.
- **Mobile Application Development**: Develop a mobile version of the application to make it accessible for doctors and healthcare professionals on-the-go, allowing them to upload and analyze images from mobile devices.
- **Multi-language Support**: Implement multi-language support to make the application accessible to non-English speaking healthcare professionals, facilitating its adoption in international healthcare environments.
- **Patient Data Management**: Integrate a secure patient data management system, compliant with healthcare regulations such as HIPAA or GDPR, ensuring that patient information and medical results are handled confidentially and securely.
- **Real-time Model Updates**: Enable real-time model updates using continuous learning techniques, allowing the model to improve its performance based on new data without the need for complete retraining.

6.3 Conclusion

In conclusion, this project has successfully demonstrated the potential of deep learning models, specifically Convolutional Neural Networks (CNNs), in the early detection and classification of brain tumors using MRI images. By integrating a pre-trained model into a Django-based web application, we have provided a practical and user-friendly tool for healthcare professionals. This system allows for quick and accurate tumor detection, which can greatly assist in improving diagnostic accuracy and speed.

Through rigorous testing and model evaluation, the application has shown reliable performance and scalability, making it suitable for deployment in real-world medical environments. The use of data augmentation, model optimization, and deployment techniques has ensured that the application is both robust and flexible for further enhancements.

Future improvements will focus on expanding the capabilities of the system to include additional tumor types, other medical imaging modalities like CT scans, and the integration of advanced features such as a medical chatbot and patient data management. These enhancements will make the tool even more valuable to medical professionals and contribute to the ongoing digital transformation in healthcare.

Overall, this project has provided a strong foundation for future work in the field of medical imaging, with the potential to significantly impact patient care and diagnostic procedures in neurology and beyond.